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SHIP TRANSPORT OF CO_2 Technical Solutions and Analysis of Costs, Energy Utilization, Exergy Efficiency and CO_2 Emissions

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Increased focus on reducing CO_2 emissions has created growing interest in CO_2 capturing from industrial processes for storage in geologic formations or injection in oil reservoirs for enhanced oil recovery (EOR). Due to the scattered CO_2 sources and the uncertainty in the growth of the CO_2 market, a cost effective and flexible transport system is required. In this work a ship transport concept is developed as an alternative to pipeline transport. New technical solutions, cost-, energy-, exergy- and CO_2 emission analysis for ship-based transport of CO_2 are presented. The concept includes all the elements in the transport chain, namely liquefaction, intermediate storage, loading system, semi-pressurized ship and offshore unloading system.

Economical large-scale transport of CO_2 by ship could be done in semi-pressurized vessels of around 20 000 m^3 at pressures near triple point (6.5 bara and $-52^{\circ}C$) in order to use well established design for commercial construction of LPG carriers and intermediate storage. This condition also gives the highest density in the liquid state, which reduces the transport unit cost. Liquefaction of CO_2 is best achieved in an open cycle, where the refrigeration is partly or fully provided by the feed gas itself. The offshore unloading system will transport the liquid CO_2 from the dedicated CO_2 ship to the wellhead on the platform at the required temperature and pressure. During the unloading phase the ship is connected to a submerged turret loading (STL) system. The CO_2 is pumped to a pressure high enough to avoid phase transition in the transfer lines. A flexible riser, a subsea pipeline and an insulated pipeline in the platform shaft bring the CO_2 from the unloading location to the topside of the platform. The CO_2 is pumped to injection pressure and heated to avoid operational problems before it is injected into the reservoir for EOR using conventional water injection wells.

The total specific energy requirement for the selected transport chain is 142 kWh $tonne^{-1}$ CO_2, where the liquefaction process accounts for 77%. An exergy analysis of the chain is performed showing that the minimum work required in the chain is 60 kWh $tonne^{-1}$ CO_2, giving a chain rational efficiency of 42%. The total CO_2 emissions are estimated to be approximately 1.4% of the inlet CO_2. The total costs of ship-based transport are calculated to be 20–30 USD $tonne^{-1}$ for volumes larger than 2 Mt y^{-1} and distances limited to the North Sea. Ship transport offers a flexible alternative for bringing CO_2 to offshore installations. Dedicated CO_2 carriers for transport of CO_2 directly from the source to the oil fields might be a key element in future CO_2 infrastructures.

Keywords: CO_2; carbon dioxide; transport; ship; costs; energy; exergy.

INTRODUCTION

Growing interest in CO_2 capturing from industrial processes and storage in underground formations is emerging from commitments on reducing CO_2 emissions manifested in the Kyoto Protocol. Among the long-term storage options of CO_2 such as injection in aquifers or other geologic storage formations, enhanced oil recovery (EOR) is the solution that can introduce an infrastructure for CO_2 handling with a financially sound basis (Hendriks *et al.*, 2002; Berger *et al.*, 2004). CO_2 transport is the necessary link between CO_2 capture and storage. The present focus on safety, flexibility, reliability, and cost reductions has created a demand for new solutions that can compete with pipeline transport (IEA, 2003).

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The most suitable sources of CO_2 are the ones with relatively high purity (>90%), e.g., ammonia plants and refineries. They cause between 0.5–1 million tonnes (Mt) of CO_2 production yearly. As several sources of the present size would be required to meet the capacity of a single reservoir, ship transport of CO_2 provides a more flexible, and in many cases, more cost-effective transport solution than pipelines.

Liquid CO_2 is produced several places in the world, mainly for the food industry. The European market is approaching 3 Mt p.a. The largest CO_2 plants reported are 0.09 Mt p.a, which is 10% of the investigated size in this work. The CO_2 is compressed to transport pressure, cleaned of unwanted components, dried and liquefied. Yara (former Norsk Hydro Agri) is presently operating four small-scale semi-pressurized CO_2 ships. These vessels are dedicated for shipping liquid CO_2 to distribution terminals in a number of countries around the North Sea basin. The vessel sizes vary between 1000 and 1500 m^3, and the transport pressure is about 14–20 bara (Hegerland *et al.*, 2004). These ships are not suitable for large-scale ship-based transport of CO_2. Lower pressure is required for enlarged storage and ship tanks, which in turn requires new processes. CO_2 is used for EOR several places onshore in the United States (Heggum *et al.*, 2005). Although the potential for increased oil recovery is likely, CO_2 is currently not used for EOR in any offshore oilfield. Offshore unloading of CO_2 or any other cryogenic liquid has not been used in any other process, and involves several new challenges.

Large-scale ship-based transport of CO_2 is relatively new, and the most important other work was presented at the 7th International Conference on Greenhouse Gas Technologies in Vancouver, 2004 (Berger *et al.*, 2004; Hegerland *et al.*, 2004; Ozaki *et al.*, 2004; Aspelund *et al.*, 2004a, b; Barrio *et al.*, 2004). Svensson *et al.* (2003) describe technical solutions and costs of ship transport of CO_2 among other transport opportunities. But, liquefaction and offshore unloading are not included in this study and economic parameters are different. Hence the calculated costs are considerably lower than the figures presented in this paper. Pipelines were already discussed by Skovholt (1993).

This paper presents innovative technical solutions for ship-based transport of CO_2, together with an analysis of the the costs, energy utilization, exergy efficiency and CO_2 emissions. All elements in the transport chain, namely liquefaction, storage, dedicated CO_2 ships and offshore unloading systems are included in the study. Capture and storage are not defined as a part of the transport chain. The CO_2 will be used for EOR, and hence decreasing the global CO_2 emissions by using the CO_2 for added value. A logistics model is applied to find the most cost-effective alternative considering the source selection, storage size ship size and transport logistics.

METHOD

Existing technology for large-scale ship-based transport of CO_2 has been investigated. Due to limited information, transport of CO_2 is compared with technologies for transport of other liquefied gases. Liquefied petroleum gas (LPG) is found to have many similarities with CO_2. Hence, for the design of the ship, tank farm, and onshore loading system, existing technology for the transport of LPG has been the starting point for the development of this innovative CO_2 technology. Novel concepts have been developed for the offshore unloading process and the liquefaction plant.

Engineering solutions for the CO_2 transport chain have been worked out supported by process and logistics modelling tools. After a feasibility study, solutions have been evaluated on a conceptual basis. Proven technology or technologies using known equipment are used where possible. Cost estimates for the transport chain have been developed based on budget prices from equipment vendors and shipyards. In order to reduce the total transport costs, energy and cost-intensive parts of the chain are detected and effort has been made to reduce the costs in these areas. As a result a logistics model has been made to fully utilize the capital-intensive ship and storage facilities.

Sensitivity analyses of transported volume, distance, and feed gas composition have been used to validate alternative modes of operation. Also hazard identification (HAZID) and a preliminary hazard and operational analyses (HAZOP) have been done in order to reduce uncertainties in the new concepts. CO_2 is heavier than air and may displace air in low areas if there is a leakage. Care must be taken when designing large-scale CO_2 liquefaction systems and storage tanks, especially in harbour areas, and a gas detector system is required. Procedures for loading and unloading liquid CO_2 near the TP are developed to avoid dry ice formation. Dry ice may form during pressure letdown in the offshore transfer system (Aspelund *et al.*, 2004a).

The processes are calculated using the commercially available simulation tool ProVision. The Soave Redlich Kwong (SRK) equation of state with a modified binary coefficient of 0.193 in the van der Waals mixing rule for water and CO_2 is used (Heggum *et al.*, 2005). The energy- and CO_2 analyses are derived from the simulation models and simple calculations.

Neglecting the inlet and outlet velocity and height and with no reactions or mixing the specific exergy change in a stream j through a unit can be written as (Kotas, 1995):

$$\Delta\varepsilon_j = (h_{Inlet} - h_{Outlet})_j - T_0(s_{Inlet} - s_{Outlet})_j \quad (1)$$

where ε is the specific exergy, h is the specific enthalpy, T_0 is the ambient temperature and s is the specific entropy. Exergy calculations are performed for the chain elements shown in Figure 1. The calculations are based on pure CO_2, with enthalpy and entropy data from (NIST, 2005). The rational efficiency describes the fraction of the input exergy in a process that is transferred to available energy, exergy. For the investigated processes the rational efficiency is:

$$\psi = \frac{\Delta E}{E_{Input}} = \frac{\sum_j m_j[(h_{Inlet} - h_{Outlet}) - T_0(s_{Inlet} - s_{Outlet})]}{E_{Electric} + E_{Fuel-oil}} \quad (2)$$

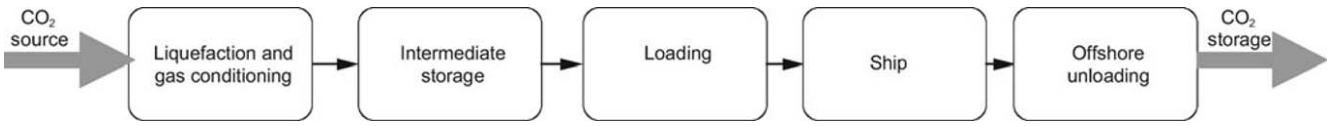


Figure 1. Main processes in the transport system.

where E_{Input} is the total required exergy, $E_{Electric}$ is the required electric energy for the sub system and $E_{Fuel\text{-}oil}$ is the required fuel oil exergy for the ship transport.

THE CO_2 TRANSPORT CONCEPT

The transport chain considered in this project comprises five main processes as shown in Figure 1: Liquefaction and gas conditioning, intermediate storage, loading, ship and offshore unloading.

To be able to liquefy the CO_2 mixture it must contain sufficient amounts of CO_2. CO_2 sources that fulfil such requirements are CO_2 from natural gas sweetening, ammonia and hydrogen production, and power plants after CO_2 capturing. However, the purity and composition varies from these sources and will affect both investment and operational costs for the liquefaction process. The CO_2 source is assumed to deliver CO_2 at 1.1 bar and 15°C. The gas is saturated with water and contains 0.3 mol% of volatile gases, 0.2 mol% of heavy hydrocarbons. Impurities influence the CO_2 properties, such as density, dew point, and boiling point, and thereby the energy requirement of the transport system.

Optimal Transport Conditions of CO_2

In order to transport large amounts of CO_2 efficiently, the gas must be transformed into a form with higher density. CO_2 can be transported as liquid, solid or in supercritical phase. Figure 2 shows the various methods for CO_2 transport and how they relate to CO_2 densities as a function of temperature and pressure (NIST, 2005).

In fully refrigerated conventional LPG and LNG ships the cargo is kept in liquid phase at atmospheric pressure by refrigeration only. At first sight this is also a logic choice for ship-based transport of CO_2. CO_2, however, cannot exist as a liquid at atmospheric pressure due to its triple point at 5.2 bar and −56.6°C. At lower pressures or temperatures CO_2 will exist either as vapour or in solid state as dry ice. The sublimation point of CO_2 at atmospheric pressure is −78°C. The density of solid CO_2 is, in theory, approximately 1500 kg m^{-3}. Solid CO_2 can be transported as dry cargo on ships, but seems not economically feasible mainly due to complex loading and unloading procedures.

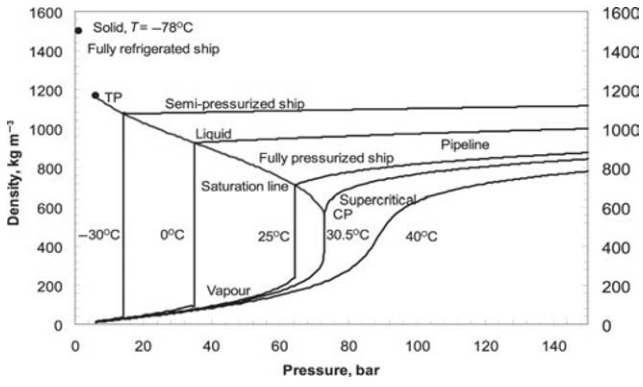


Figure 2. Optimal transport pressure.

In semi-pressurized ships the gas to be transported is kept in liquid phase on the saturation line by a pressure higher than atmospheric pressure and a temperature lower than ambient temperature. CO_2 exists in liquid form at pressures between 5.2 bar, triple point (TP) and 73 bar, critical point (CP). The density of saturated liquid will range from 1200 kg m^{-3} at the TP, to 600 at the CP. So in a fixed volume vessel almost two times more CO_2 can be transported at low pressures near the TP than at high pressures near the CP. Moreover, the lowest pressure possible is desirable in order to keep the production cost of the pressure vessels on the ship as low as possible. The required energy to liquefy CO_2 increases with decreasing pressure, however, the marginal penalties for the decreased pressure is small compared to the benefits of higher transport capacities and less costly vessels. It is likely that the CO_2 will be transported at a pressure near the triple point (TP) although operating near the TP can lead to operational problems. When the pressure is reduced below the TP equilibrium dry ice will not necessarily form immediately, as a certain sub-cooling is required for the first nucleons of dry ice to form. When sufficient nucleons are made the dry ice will form rapidly. So there is an additional buffer for dry ice to form, but when it starts to freeze out it is a rapid process where a lot of dry ice is formed in seconds. Care must be taken to avoid dry ice formation in the process, storage and loading/unloading since blockage and operational problems may occur.

Liquefaction and Gas Conditioning

The liquefaction plant delivers CO_2 at 6.5 bar and −52°C to the storage tanks. The capacity of the reference liquefaction plant is 1.0 Mt CO_2 p.a. however, the same technology can most likely be used in plants between at least 0.5 and 2.5 Mt CO_2 p.a. The plant has access to seawater at 10°C. The minimum internal temperature approach in the ambient heat exchangers are 5°C.

In state-of-the-art technology the CO_2 is compressed to transport pressure of 14–20 bar, before it is liquefied by an external refrigeration cycle, using ammonia as refrigerant. Hence, the process that is most similar to the state-of-the-art is to compress the CO_2 to transport pressure of 6.5 bar and liquefy it using and an external refrigeration cycle. This solution is, however, not economical preferable as cryogenic refrigeration at −50°C is complex and the energy efficiency is low due to exergy losses in the heat

exchangers at low temperatures. Also, more water must be removed in the drying unit since less water can be removed in the gas scrubbers at the low pressure. The lowest preferable pressure for condensation of CO_2 is 15–20 bar, which fully utilize a two-stage centrifugal compressor.

Large-scale liquefaction of CO_2 is best achieved in an open cycle, using the CO_2 feed as refrigerant where the refrigeration is partly or fully provided by the feed gas itself. The CO_2 feed is first compressed in stages to a pressure higher than the transport pressure. The CO_2 is cooled by ambient air or water after each compressor stage. At the highest pressure heat is rejected from the CO_2 in one of three configurations:

- C1, heat rejection from the CO_2 to an external cooling circuit, at 20 bar (−15°C), Figure 3;
- C2, heat rejection from the CO_2 to the ambient (seawater), at 55 bar (15°C), below the critical point, Figure 4;
- C3, heat rejection from the CO_2 to the ambient (seawater or air), at 95 bar (15°C), over the critical point, Figure 5.

The liquid or supercritical CO_2 is sub-cooled by flash gas from the expansion or liquid CO_2 at intermediate pressures. Subcooling will improve the process efficiency significantly. The liquid CO_2 is then expanded in one or more stages to reach the liquid product specifications. Liquid expanders can be used in order to increase the process energy efficiency, but is not necessarily cost-effective. The flash gas from the expansion and boil-off gas from the storage tanks is sent back to the compressor train at the appropriate pressure level for re-compression.

Water must be removed to avoid hydrates, freezing of water and corrosion. The solubility of water in CO_2 gas decreases with higher pressure and lower temperatures. Therefore, most of the water is removed in gas scrubbers after compression and cooling (points A in Figures 3–5). The last free water is removed at a pressure between 20 and 40 bar and at a temperature close to the hydrate formation curve. The CO_2 gas is dried to ppm level by adsorption (points B in Figures 3–5). The purge gas is integrated in the process to avoid loss of CO_2.

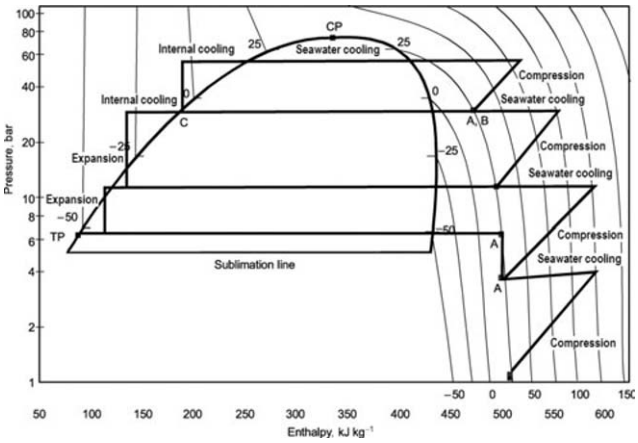


Figure 4. C2, condensation against seawater.

The removal of volatile components is done in a packed separation column after heat rejection close to the dew point line, (points C in Figures 3–5). At transport pressures of 6–7 bar only small fractions of volatiles, typically 0.2–0.5 mole% can be included in the product to insure that dry ice is not formed. If more volatiles are present in the feed they must be removed. A column is used to avoid venting of large quantities of CO_2 to the atmosphere. The cooling duty in the condenser is provided by vaporization of liquid CO_2 at intermediate pressure stages or from the product tank. As a rule of thumb the loss of CO_2 will be equal to the amount of volatiles in the feed.

Centrifugal compressors will be used for compression of CO_2. The number of compressor stages will depend on inlet pressure and at what pressure heat is rejected from the CO_2. C1 will most likely use a two-stage compressor as shown in Figure 3. C2 will require at least three stages, with four stages as the most likely solution, Figure 4. C3 will need four stages of compression, Figure 5. A detailed description of the liquefaction process is found in the international patent application 'Process and Apparatus for Liquefaction of Carbon Dioxide' WO2006008482 (Aspelund *et al.*, 2006a).

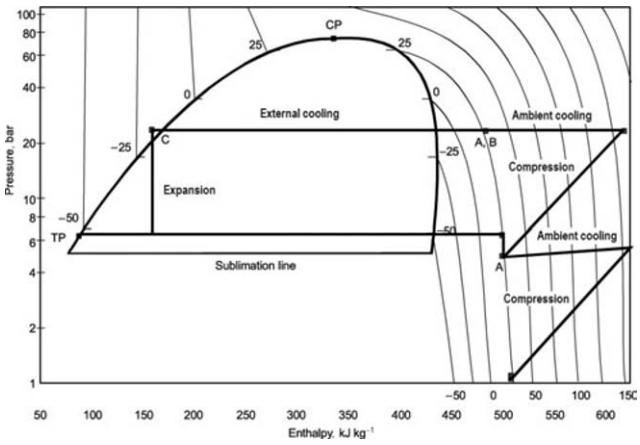


Figure 3. C1, condensation by external cooling.

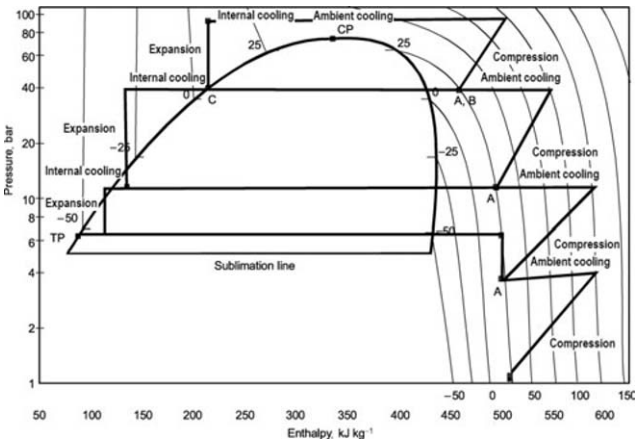


Figure 5. C3, cooling at CO_2 over the CP.

Intermediate Storage

The CO_2 is stored at the bubble point in semi-pressurized storage tanks until the ship berths at quay. The design of the storage facility has to be flexible regarding size, place and ground conditions as the need for storage volume, and variations in geographical location. The total storage capacity is set to 30 000 m^3 (1.5 times the ship capacity). Semi-pressurized storage is common for other liquefied gases as LPG and ethylene. The most common method is using semi-pressurized spheres, semi-pressurized cylindrical tanks or underground storage in caverns. The pressure vessels can be placed above the ground on racks or they can be placed on the ground where they are partly or fully covered.

A generally applicable solution is chosen. The tank farm consists of 10 cylindrical steel storage tanks placed horizontally. The volumes of the tank farm are in the upper end of existing storage facilities, however, tank farms with capacities in excess of 50 000 m^3 in 30 or more pressure tanks exist (McGurie and White, 2000).

Loading System

The loading system at quay transfers liquefied CO_2 from the storage tanks at the liquefaction plant to the ship. The loading system includes all the necessary piping between tanks and ship, as well as pumps, marine loading arm and export building. The recommended solution consists of two parallel product pipes between the tanks and the loading arm for export of CO_2 and a return line for CO_2-vapour generated at the ship.

The loading system equipment and materials have been carefully chosen by looking at their resistance to corrosion and low temperature, the need for maintenance, price and availability. Also procedures for loading are developed to prevent operational problems with dry-ice formation or problems with thermal cycling.

Ship

Semi-pressurized ships are usually designed for a working pressure of 5–7 bara and operate at low temperatures (-48°C for LPG, -104°C for ethylene). This is the most frequent type of ship for LPG transport up to nearly 20 000 m^3. Such vessels have normally two to six tanks, and each tank may have a capacity of 4500 m^3. It may be possible to rebuild existing LPG ships for CO_2 transport. However, since CO_2 is transported at slightly higher pressures, higher densities and lower temperatures than LPG, some difficulties may arise. Only a few of the existing LPG ships can withstand the CO_2 design requirements. The design of a tank ship is dictated to a large extent by the international maritime regulations, class regulation as well as the conventional ship design parameters that control drag, manoeuvrability, weight, stability, freeboard and relative strength. In addition, the shipyard production line may affect the dimensions of the ship.

The concept suggested is a combined LPG/CO_2 semi-refrigerated ship with a total capacity of 20 000 m^3, adapted to both submerged turret loading (STL) unloading offshore and direct terminal/terminal service. The reasons for choosing a combined LPG/CO_2 ship are increased remaining value, and the possibility of improving ship utilization. The need for re-condensation equipment on board depends on whether the ship will also be used for LPG transport and on the distances for CO_2 transport. CO_2 transport within the North Sea does in principle not require re-condensation on board. The ship is equipped with the necessary elements (heat exchanger, pump, and so on) so that CO_2 can be heated to 0°C previous to unloading offshore. Figure 6 shows a typical arrangement for a CO_2 ship.

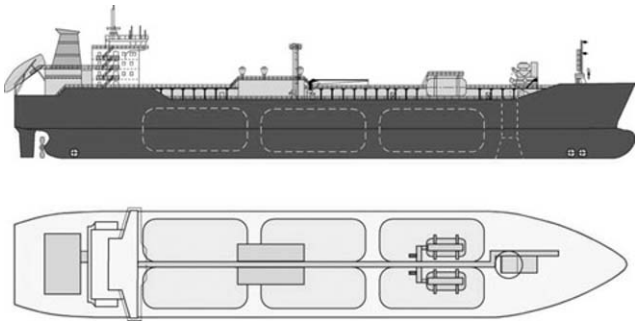


Figure 6. Typical arrangements for a CO_2 ship.

Offshore Unloading System

The CO_2 will arrive at the oil field in the ships at 6.5 bar and -52°C. In order to overcome the reservoir pressure and avoid hydrate and wax formation the inlet conditions to the wellhead will be at 200–300 bar and 15–20°C. The offshore unloading system will be connected to the platform via a riser and a seabed pipeline for transfer of the CO_2, and must be able to keep the vessel at a safe distance from the platform. The availability of the unloading system is of high importance to ensure a cost-effective transport chain. Furthermore a high unloading rate is important to increase the ship capacity and reduce the total transport costs.

The offshore unloading system shown in Figure 7, basically includes:

- a submerged turret loading (STL) system;
- a submerged cargo pump in the cargo tank and a booster pump on the ship deck;
- a CO_2 heating system on the ship and/or a waste heat recovery heating system at the platform;
- a flexible riser from the ship to the seabed;
- a pipeline at seabed;
- a pipeline from seabed to platform deck;
- a pumping system on the platform for pressure increase up to injection pressure;
- an expansion system on the platform to avoid pressure build-up in the transfer lines between transmission.

In the most flexible solution the CO_2 is pumped to a pressure somewhat higher than the injection pressure, before it is heated to ambient temperature and transferred to the injection manifold on the platform for re-injection. This solution is valid for most offshore installations. In warm waters the heating can be provided by heat exchangers using seawater as heating media. In the North Sea, with temperatures as low as 5°C, heating with seawater is not technically preferable due to icing in the heat

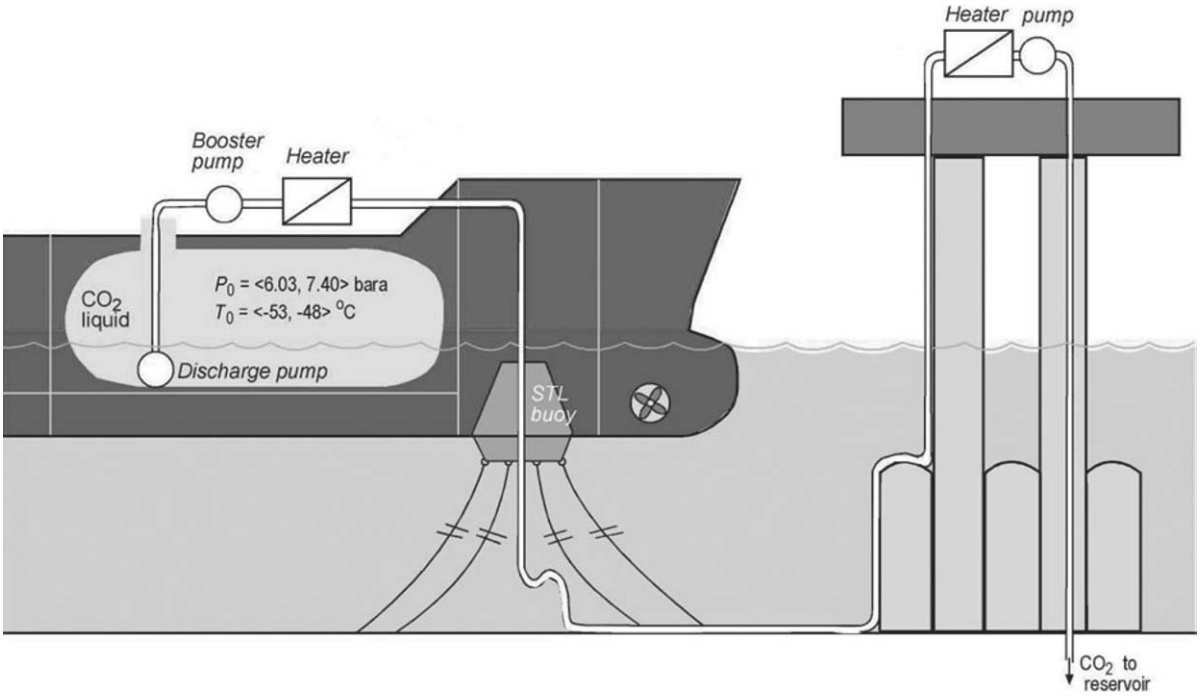


Figure 7. Sketch of submerged turret loading (STL) system used for unloading of CO_2.

exchangers. Therefore, heating of the CO_2 will require a heating system, which will increase both the capital and operational costs. Moreover, there will be a significant increase in CO_2 emissions due to burning of fuel for the heating process. To avoid high-pressure transfer lines, pumping from 60 bar to the required injection pressure may take place on the platform.

If waste heat and sufficient space for an additional heat exchanger and pumps are available on the platform, the CO_2 should be heated on board the platform. The liquid CO_2 is pumped by the booster pump to 50–60 bar in order to avoid flashing of CO_2 in the transfer system. The cold CO_2 is unloaded through the transfer system at $-45^{\circ}C$. On the platform the CO_2 is heated to $10-20^{\circ}C$ by waste heat before it is pumped to 200–300 bar. The main advantage is the utilization of excess waste heat on the platform. At as low temperatures as $-50^{\circ}C$ many materials become brittle. Hence, it is a challenge to design a flexible riser for such cold liquids. Icing in the STL and on the flexible riser may also cause problems if not designed for. Moreover, ice formation on the subsea transfer lines will act as an insulating layer and reduce the heat transfer to the pipeline significantly. Also depressurization of the transfer line is a challenge when unloading at low pressures and temperatures. The ship is an important part of the total costs in the transport chain and an efficient utilization is essential. Therefore continuous unloading from the ship will be a too costly solution. Technically an intermediate storage of CO_2 could be a possibility, however this is also a too cost-intensive. Hence, in order to keep the unloading time at a minimum, the CO_2 will be injected in batches at high rates while the CO_2 ship is connected to the transfer system. Between the batches of CO_2 produced water may be injected. Batchwise injection of CO_2, water and natural gas may actually be favourable, from a reservoir point of view, compared to continuous injection from a pipeline. Also the existing water injection wells may be utilized resulting in reduced investment costs. To avoid depressurization of the unloading system between each discharge operation, the system is kept at transmission pressure. As the cold liquid CO_2 is heated from the surrounding seawater the density of the CO_2 will decrease and hence the volume will expand. To avoid venting of CO_2 to the atmosphere an expansion system, which accumulates liquid CO_2 in the period between each discharging operation is included. A detailed description of the offshore unloading system is found in the international patent application WO2006008486, 'Vessel' (Aspelund *et al.*, 2006b).

ANALYSIS OF THE CO_2 TRANSPORT CONCEPT

Cost Estimate

The cost estimate is based on investment costs provided by several equipment suppliers. Total costs are calculated using an interest rate of 7%, and an economic life of 15 years. The remaining value for the ship is set to 35% of the investment, for all other installations it is set to zero. All prices are given in 2003 USD tonne^{-1} CO_2. Exchange rates are set to 7.0 USD NOK^{-1} and to 7.8 EUR NOK^{-1}, and the electricity price is set to 0.03 USD kWh^{-1}. Zero emissions of CO_2 during transport are assumed. CO_2 is assumed to be freely available at the source at no cost (Table 1).

A cost estimate for ship transport of CO_2 from two real sources located at Antwerp and Brunsbüttel to a given end-user offshore has been performed. A roundtrip distance approximately 1500 km is used in the calculations. The total costs for transport of CO_2 for this specific case are

Table 1. Logistics data.

Harbour	Antwerp	Brunsbüttel
CO_2 delivered (tonne year^{-1})	644 000	1 288 000
Storage capacity (m^3)	21 000	30 000
Approach/departure/mooring time (h)	19	12
Loading time (h)	13	13
Harbour tax (USD)	39 300	31 488
Round trip to offshore end-user (h)	68	49
Number of calls per year	28	56

calculated to 25 USD tonne^{-1}. Investment, operational, and capital costs are included in the total costs. Table 2 shows the costs for each part system of the transport chain. In Figure 8 total costs are given for transport from various collections of real sources in Northern Europe to a certain offshore end-user within the North Sea. The figure illustrates that the transport costs are in the range from 20–30 USD tonne^{-1} CO_2 for volumes larger than 2 Mt CO_2 y^{-1}. The line in the figure shows the trend of reducing costs at larger volumes.

Energy Analysis

An energy analysis for each subsystem of the transport chain is performed. The energy requirements for the liquefaction processes are retrieved from the simulation models. Five percent is added for losses in the electric drives and utilities. The process C2 is used in the calculations. The energy requirement for the ship is calculated from an average consumption of 0.6 tonne h^{-1} of fuel oil with a heating value of 41 MJ kg^{-1} and a total transport time of 7000 h. The total energy requirement is 117 kWh electric power and 2.17 kg (25 kWh) oil per tonne CO_2. The results are presented in Table 2.

The liquefaction and conditioning system is the most energy-demanding process and is therefore analysed in detail. The energy requirement depends on ambient temperature, inlet pressure and composition.

As the inlet pressure increases the energy requirement decreases rapidly at first and slower as the pressure increases. This is expected as the same compressor duty varies with pressure ratio, not absolute pressure. The energy requirement for process C2 at increasing inlet pressure is 110, 89, 67, 40 kWh tonne^{-1} CO_2 for the inlet pressures 1, 2, 5 and 20 bar, respectively. The efficiency of the liquefaction process is also dependent on the compressor efficiency. One percent change in polytropic efficiency will change the liquefaction energy requirement with 2 kWh tonne^{-1} CO_2.

Figure 9 shows the required liquefaction energy as a function of the final heat rejection temperature for the three evaluated configurations. C2 will have an increase of 2 kWh tonne^{-1} CO_2 per °C, while the increase for C1 and C3 is 1 kWh tonne^{-1} CO_2 per °C. The reason for this is that C2 operates in the two-phase area close to the CP, which decreases the process efficiency rapidly. C2 should not be used for higher condenser temperatures than 25°C and is therefore limited to seawater cooling only. The intersection point between C1 and C2 is at a condensing temperature of 20°C (seawater temperature of 15°C). If ambient air or warmer seawater than 15–20°C is used configuration C1 with an external cooling unit should be used. It is possible to avoid the external cooling unit even at high ambient temperatures, by compressing the CO_2 to a supercritical pressure and remove heat by cooling of supercritical CO_2. This configuration, C3, has the highest power consumption.

Figure 10 shows the required liquefaction energy as a function of volatile components in the feed. The effect of impurities in CO_2 transport processes is discussed in an article describing the interface between CO_2 capture and transport (Aspelund and Jordal, 2006). As can be seen the three configurations all have an increase of 4 kWh tonne−1 CO_2 per mole % nitrogen.

Exergy Analysis

An exergy analysis will, from a thermodynamically point of view; provide information of where in the transport chain there are unnecessary losses. The total chain rational efficiency is close to 40%. The results are presented in Table 2.

CO_2 Emissions

Most capture processes will not capture all the CO_2 as the marginal energy and investment costs are high for last percents of the CO_2. Also, in a CO_2 transport chain some CO_2 will be lost during gas treatment and conditioning. Some CO_2 will be emitted from the ship engines and from the gas turbines providing energy to the offshore process. Therefore CO_2 capture and injection will never be 100% effective.

In the calculations the 110 kWh tonne^{-1} CO_2 needed in the liquefaction storage and loading subsystem is taken from a power plant with 100% CO_2 capture. Hence there are no CO_2 penalties for the required energy in these processes. However, if the power is taken from a state-of-the-art combined cycle, CC, power plant which emits 0.35 kg CO_2 kWh^{-1}, a loss of 3.85% of the captured CO_2

Table 2. Summary of the CO_2 transport concept analysis.

	Liquefaction	Storage	Loading	Ship	Unloading	Total chain
CO_2 inlet phase	V	L	L	L	D	
Inlet P (bar)	1.1	6.5	6.5	6.5	200	
Inlet T (°C)	15	−52	−52	−52	15	
Specific energy requirements (kWh tonne^{-1})	110	0	0.2	25 (Oil)	6.5	**142**
Specific exergy requirements (kWh tonne^{-1})	60	0	0	0	0.1	**60**
Rational efficiency (%)	55	0	0	0	1.5	**42**
Relative CO_2 emissions (%)	0.3	0	0	0.65	0.45	**1.4**
Specific costs (2003 USD)	10.5	2.2	0.8	7.5	4	**25**

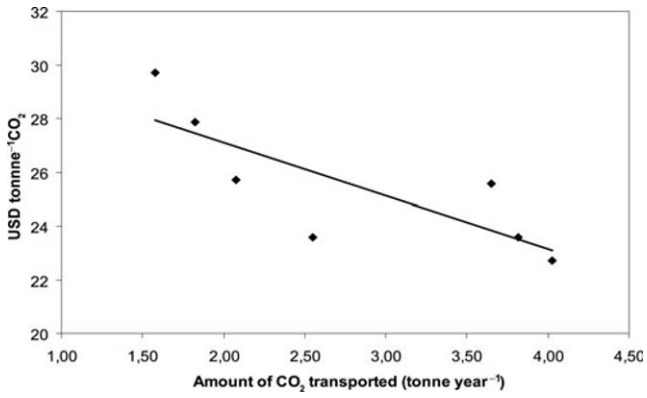


Figure 8. The total costs for transport from real sources considered in the logistics calculation. The sailing distances are limited to the North Sea.

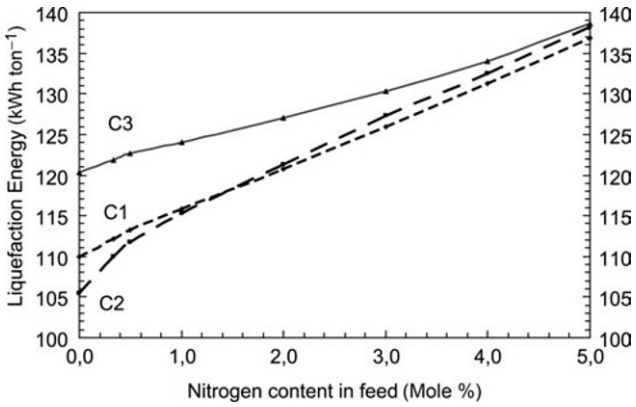


Figure 10. Energy requirements as a function of nitrogen content in feed.

can be expected. If 90% is captured in the power process, the loss would be 0.39%. The boil-off gas from storage and loading is returned to the liquefaction unit, hence there are no CO_2 emissions from these systems. In the liquefaction unit there will be minor losses of CO_2 in the waste water as some CO_2 is solved in the water. If there are volatile components in the feed gas they must be removed. The molar loss of CO_2 will be approximately equal to the amount of volatiles in the feed unless the volatile purge stream is pretreated or recycled to the industrial or capture process.

The ship and offshore unloading system will not emit any of the captured CO_2 to the atmosphere as the tank pressure is allowed to increase to account for the heat leak from the ambient; however, the ship engines will emit 1.86 tonne CO_2 per hour, corresponding to 0.65% of the transported CO_2. At the offshore unloading facilities the required energy for pumping of CO_2 is provided by a gas turbine with efficiency of 30%, giving a CO_2 emission of 0.45% of the transported CO_2. The results are presented in Table 2.

SUMMARY OF THE CO_2 TRANSPORT CONCEPT ANALYSIS

DISCUSSION

The liquefaction plant accounts for 77% of the specific energy requirement and 43% of the specific costs and is therefore the most important system in the chain. The costs for liquefaction are dominated by the investment and operation of the compressor train. The rational efficiency of the liquefaction plant is high, 55%, hence it is not possible to increase the efficiency much unless it is integrated with the capture process or other industrial processes, such as vaporization of liquefied natural gas (LNG). An increase in the inlet pressure will decrease the energy requirements significantly, therefore the CO_2 should be captured at as high pressure as possible. For large quantities of volatiles in the CO_2 the purge gas should be recycled to the capture or industrial process to avoid emitting equally large amounts of CO_2 to the atmosphere.

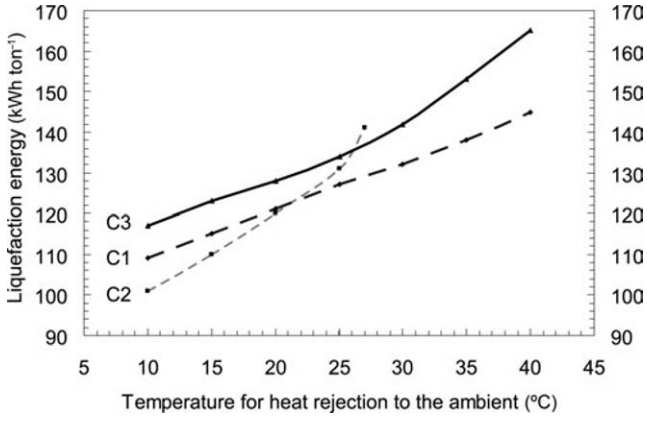


Figure 9. Energy requirements as a function of heat rejection temperatures.

A well-designed and optimised storage and unloading system will have small energy requirements and only account for 12% of the total costs. Furthermore, it is based on proven technology. The suggested storage is quite space demanding, hence for large volumes, other technologies like gas caverns might be a better solution.

The ship accounts for 19% of the specific energy requirement and 30% of the specific costs and is the second most important part of the chain. For the given distance the ship is the largest emitter of CO_2. Efficient utilization of the ship capacity is in therefore very important for minimizing the costs, CO_2 emissions and energy requirement. If the fuel is changed from bunkers oil to natural gas the emissions will be reduced with approximately 30%. The NO_x emissions will also be reduced significantly.

The exergy in the liquid CO_2 at the ship and the high pressure CO_2 in the dense phase to be injected are approximately the same, giving a rational efficiency in offshore unloading close to zero. This means that the offshore energy requirement of 6.5 kWh tonne^{-1} is lost unless it is used for process cooling on the offshore installation. Due to the batch unloading of CO_2 one offshore unloading system can serve several ships, thereby reducing the specific costs of the system.

CONCLUSIONS

Ship transport offers a cost-effective and flexible alternative to pipeline transport for bringing CO_2 to offshore installations. Dedicated CO_2 carriers for transport of CO_2 directly from the source to the oilfields might be a key element in future CO_2 infrastructures. New technical

solutions for ship-based transport of CO_2 are developed. The concept includes liquefaction, intermediate storage, loading system, semi-pressurized ship and offshore unloading system.

Economic large-scale transport of CO_2 by ship should be done in semi-pressurized vessels at pressures near the triple point e.g., at 6.5 bar and $-52^{\circ}C$. Liquefaction of CO_2 is best achieved in an open cycle, using the CO_2 feed as the refrigerant. Storage and loading system are similar to the LPG concepts. The ship concept suggested is a combined LPG/CO_2 semi-refrigerated ship with a total capacity of 20 000 m^3, adapted to both submerged turret loading (STL) unloading offshore and direct terminal to terminal service. The offshore unloading system will be connected to the platform via a riser and a seabed pipeline for transfer of the CO_2.

The total specific energy requirement for the selected transport chain is 142 kWh $tonne^{-1}$ where the liquefaction process accounts for 77%. An exergy analysis of the chain is performed showing that the minimum work required in the chain is 60 kWh $tonne^{-1}$ CO_2, giving a chain rational efficiency of 42%. The total CO_2 emissions are estimated to be approximately 1.4% of the inlet CO_2. The total costs of ship-based transport are calculated to 20–30 USD $tonne^{-1}$ for volumes larger than 2 Mt y^{-1} and distances limited to the North Sea. For a selected case including sources in Antwerp and Brunsbüttel of together 2 Mt y^{-1}, the total costs are calculated to 25 USD $tonne^{-1}$.

NOMENCLATURE

CO_2	carbon dioxide
CP	critical point
EOR	enhanced oil recovery
LPG	liquefied petroleum gas
LNG	liquefied natural gas
Mt	million tonnes
TP	tripple point
STL	submerged turret loading

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